

An Urban Electric Vehicle Charging System via Hybrid Heterogeneous Modes

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Abstract—Electric Vehicle (EV) is regarded as the optimal alternative to traditional fuel-powered vehicles. However, the exponential surge in EV charging demand poses challenges in charging infrastructure planning and charging behavior management. The efficacy of traditional Grid-to-Vehicle (G2V) charging mode, which obtains power from the grid, is curtailed by the limited number and uneven distribution of charging facilities, inevitably leading to charging congestion. Instead, the concept of Vehicle-to-Vehicle (V2V) charging has emerged as a spatio-temporally flexible charging mode, which expands EV's role from consumer to provider, forming V2V pairs and utilizing urban Parking Lots (PLs) as charging locations. In this paper, we propose a Hybrid Heterogeneous Modes (HHM)-based EV charging optimization scheme in urban settings. Building upon the G2V mode, we introduce synchronous and asynchronous V2V charging modes as optimization strategies, integrating and exploiting the unique advantages of each mode. We also utilize global EV charging scheduling and comprehensively take four dimensions into consideration (energy trading cost, travel cost, waiting time cost and loss cost), thus achieving flexible mode selection, V2V pairing, and designated charging locations. Simulations confirm the effectiveness of the proposed scheme in reducing total charging costs, optimizing user service experiences, and improving charging facility utilization rates.

I. INTRODUCTION

In recent years, human activities have exacerbated global warming, with emissions from traditional fuel vehicles being a significant man-made contributor. Electric vehicle (EV), powered by onboard electricity, is widely regarded as the best alternative to fuel-powered vehicles. It helps reduce reliance on fossil fuels, curb greenhouse gas emissions, and enhance energy conversion efficiency. However, the large-scale EV adoption poses challenges to the existing power system, including unordered charging behavior, high loads during peak times, and demands surpassing service capacity. These challenges have prompted research into EV charging optimization and exploration of diverse charging modes.

Previous works mainly focus on Grid-to-Vehicle (G2V) charging mode [1]–[4], where Charging Station (CS) serves as the primary charging location, drawing power inputs from the grid or local energy generators. When charging demand emerges, EVs as energy loads proceed to CSs and get

charged through deployed charging slots. Nevertheless, with the rapid surge in charging demand, this mode faces issues. Firstly, the power system is strained by the increased burden of EV loads, leading to elevated system peaks and overloads [5]. Secondly, limited charging infrastructure, especially the scarcity and restricted distribution of CSs in urban settings [6], incurs range anxiety and drives up total charging costs.

Vehicle-to-Vehicle (V2V) charging mode is akin to the concept of Vehicle-to-Grid (V2G) [3], where EVs serve as energy providers, leveraging their capability as distributed energy storage units. Compared to G2V charging, V2V mode categorizes EVs into Consumer (EV-C) and Provider (EV-P) [7], which emphasizes on interconnecting EV pairs. Moreover, V2V energy transfer is spatio-temporally flexible, as it can occur anytime and anywhere (e.g., generally at Parking Lots (PLs)) via DC (Direct Current)-DC converters [8]. Based on timing and facilities, the V2V charging mode can be further divided into synchronous V2V and asynchronous V2V [9], [10]. The former requires matched pairs of EV-C and EV-P to be present simultaneously at the charging location for energy transfer, while the latter eliminates this requirement by utilizing energy storage facilities deployed at the charging location. Thus, matching between EV-C and EV-P is not practically significant in asynchronous V2V mode.

Charging optimization in V2V mode involves V2V-pair matching, selection of charging locations, and charging scheduling [11]. Additionally, optimization objectives vary, which may include alleviating range anxiety to enhance Quality of Experience (QoE) [12], increasing EV-P benefits from energy supply, maximizing service quantity, minimizing global costs, etc. Some studies view V2V charging mode as a supplement to G2V mode, aiming to balance grid loads, reduce infrastructure investment, and ease charging congestion when demand exceeds CS capacity [10], [11]. However, the charging rate of V2V mode remains lower than that of G2V due to hardware limitations. Despite its unique advantages in spatial and temporal domain, challenges such as margin of charging time, user participation [13], security, and privacy hinder V2V mode from completely replacing G2V mode.

Considering the shortcomings of charging modes mentioned above, we propose a Hybrid Heterogeneous Modes

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(HHM)-based EV charging optimization scheme in urban settings. The main contributions of this paper can be summarized as follows:

1. Compared to traditional G2V/V2V charging scenarios [4], [7], our proposed scheme combines three charging modes: G2V, synchronous V2V and asynchronous V2V, within the scenario. Through efficient utilization of heterogeneous charging facilities in urban settings for hybrid charging, our scheme can transfer demand to suitable charging modes based on the grid load. Flexible mode selection helps maintain the stability of the power system.
2. A charging cost function is proposed from user's perspective, which comprehensively considers four dimensions: energy trading cost, waiting time cost, travel cost, and loss cost. Our scheme applies it to evaluate potential charging plans in three modes to formulate global optimization decisions, thus minimizing total charging costs and improving user QoE.

II. RELATED WORK

EV charging management involved multiple stakeholders, including CS operators, power system operators, and EV owners. However, current research hadn't reached a consensus on benefit dimensions (optimize CS operations, coordinate urban power system management, owners' economic benefit, etc.) for these stakeholders. Centralized charging management offered advantages with respect to modeling complexity [14], so the research trend still focused on addressing centralized charging optimization within a single mode (G2V, V2V). Although the potential of the V2V mode had been recognized as investigations deepened, management schemes that integrated and coordinated multiple feasible charging modes had not received sufficient attention. There was a lack of corresponding optimization model designs and control strategy formulations for such schemes.

A. G2V Charging Mode

Numerous works focused on the charging scheduling issue in EV charging optimization, and adopted centralized or distributed approaches. The work in [3] presented a distributed scheduling algorithm to reduce operational costs by coordinating multiple CSs, while literature [4] proposed a two-stage charging scheduling algorithm (the second stage was centralized) to mitigate adverse impacts of uncontrolled charging on the grid. Literature [5] further considered the impact of CS site selection on G2V. The work addressed the siting planning of CSs in urban areas, which was designed to minimize the social cost of whole charging system. Some studies recognized EVs' potential as mobile energy storage unit, thus the concept of V2G emerged as an optimization direction, which facilitated bidirectional energy flow between EVs and the grid. Review [6] summarized research on EV charging and discharging scheduling under grid constraints.

B. Synchronous and Asynchronous V2V Charging Mode

For synchronous and asynchronous V2V modes, studies often focused on different stages of V2V charging process. To optimize V2V-pair matching, the work in [7] constructed a weighted bipartite graph to maximize user utility. Once a V2V-pair was matched, charging location should be specified. The work in [12] proposed a semi-centralized charging navigation framework, which determined the optimal PL for EVs. Then, Literature [12] designed an adaptive path selection algorithm based on Q-learning to optimize the travel process. Emphasizing energy trading process, the work in [13] optimized participants' interests via Double Auction mechanism. However, few studies extensively delve into the integration of V2V and G2V mode. Literature [11] integrated G2V and synchronous V2V charging modes, introducing reservation mechanisms and parking duration constraints.

Inspired by the mode selection between synchronous and asynchronous V2V charging in previous work [15], we redesign the scheme, integrating G2V charging mode to suit real-world scenarios. Our work elaborate on the charging rules of each mode, enabling PLs to support both synchronous and asynchronous V2V modes, while CSs support G2V mode. We also detail and distinguish the implementation logic of three modes, aiming to leverage their unique advantages. Additionally, we redefine the optimization objectives, adjust waiting time cost, and expand dimensions such as energy trading cost, travel cost and loss cost. These resolve the charging optimization issue via hybrid modes.

III. PRELIMINARY

A. Assumption

Our system is designed for urban scenario, comprising network entities such as Global Controller (GC), EV, CS, and PL. They utilize 5G communication technology for wireless communication and information exchange. To ensure reliable message transmission among entities, we assume that trust relationships have been established, while encrypted communication is employed to prevent unauthorized entities from accessing real-time location, demand, user identity, and other private information.

B. Network Entities

Global Controller (GC): The GC is responsible for centrally managing all charging services. It collects real-time information such as EV requests, EV physical locations, and occupancy status of CSs/PLs. Then it computes the charging cost to manage charging and discharging allocation of EVs, thereby facilitating global charging service optimization.

Charging Station (CS): CS serves as the charging location in G2V mode, equipped with certain charging slots. Due to high construction cost, urban space limitations, and difficulties in extending the grid system, the number of CSs is fewer than PLs. Energy prices follow the Time of Use (ToU) model proposed in [16] to simulate real-world conditions.

Parking Lot (PL): PL serves as the charging location in both synchronous and asynchronous V2V mode. For the former, it only provides spaces, assuming EVs carry onboard

DC-DC converters. For the latter, each PL is equipped with an energy storage facility (battery) to temporarily store energy transferred from EV-Ps, with a limited number of interfaces.

Electric Vehicle (EV): EVs can be classified into EV-Cs and EV-Ps. Both have State of Charge (SoC) thresholds as benchmarks for generating charging/discharging requests.

C. System Model and Charging Cost Definition

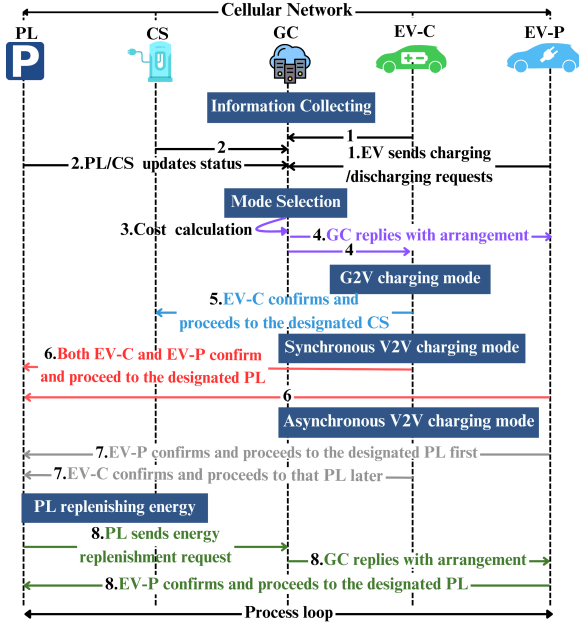


Fig. 1. Time Sequence for the System Procedure

Fig.1 illustrates the time sequence of our system (represented by the numerical order from 1 to 8). Based on collected information (Numbers 1-2), GC aims to minimize the total charging cost, thereby rationally arranging EVs' charging modes, V2V-pair matching and charging location selection (Numbers 3-4). The charging cost Γ_ψ is calculated as

$$\Gamma_\psi = \omega_e * COST_e + \omega_{tra} * COST_{tra} + \omega_{wait} * COST_{wait} + \omega_{loss} * COST_{loss} \quad (1)$$

where Γ_ψ includes four dimensions: energy trading cost ($COST_e$), travel cost ($COST_{tra}$), waiting time cost ($COST_{wait}$), and loss cost ($COST_{loss}$). ω , assigned based on the Analytic Hierarchy Process (AHP) [11], represents the weights of each dimension. It should be noted that ψ represents three charging modes, which refers to Syn (Synchronous V2V mode)/Asy (Asynchronous V2V mode)/G2V (G2V mode), respectively.

1. Energy trading cost $COST_e$ is defined as

$$COST_e = M^\psi * (E_{evc}^{req} + E_{ev}^{tra}) \quad (2)$$

where M^ψ indicates unit energy price. It directly affects the charging cost and user willingness. In G2V charging mode, we adopt the ToU pricing model from the work in [16], which sets prices M_t^{G2V} based on the energy load during a day (t represents time interval). Literature [17] assumes

that EVs charge at private residences during the night at an original price ρ , which provides a reference for determining V2V transaction prices. Additionally, considering the rental of energy storage facilities in asynchronous V2V mode, we set its price M^{Asy} higher than that of synchronous mode (M^{Syn}). E_{evc}^{req} and E_{ev}^{tra} represent EV-C's charging demand and its travel energy consumption, respectively. Their sum is the actual charging amount.

2. Travel cost $COST_{tra}$ is defined as

$$COST_{tra} = (DIS_\phi^{ev} / V_{ev}) * \gamma \quad (3)$$

where DIS_ϕ^{ev} is the distance between EV and ϕ . ϕ represents the charging location, which differs in G2V and V2V modes (refers to CS or PL). V_{ev} is the speed of EV, and we assume that all EVs in the scenario travel at a constant speed. Since E_{ev}^{tra} has been considered in equation (2), equation (3) only calculates the travel time of EV to the designated location. The time needs to be multiplied by the time value γ , which converts the time into monetary value. Through it, all dimensions can be normalized to monetary value.

3. Waiting time cost $COST_{wait}$ is defined as

$$COST_{wait} = (T_\phi^{wait} + (E_{evc}^{req} + E_{ev}^{tra}) / P_\psi^{cha}) * \gamma \quad (4)$$

where T_ϕ^{wait} (calculated in Algorithm 1) is the time from the EV's arrival (at CS/PL) to the start of charging service, then plus the actual charging time. P_ψ^{cha} represents the charging rate in mode ψ . Total time is also multiplied by γ .

4. Loss cost $COST_{loss}$ is defined as

$$COST_{loss} = Cost_{bat} * \zeta * (E_{evc}^{req} + E_{ev}^{tra}) + Brand + (1 - \delta_\psi) * (E_{evc}^{req} + E_{ev}^{tra}) * \rho \quad (5)$$

where the loss cost includes: (a) Battery cycle aging cost, which is caused by charging and discharging behavior. (b) Brand coefficient. (c) Energy loss during transmission.

Specifically, (a) is the product of battery replacement cost $Cost_{bat}$ given by [7], battery degradation coefficient ζ , and total energy transfer volume ($E_{evc}^{req} + E_{ev}^{tra}$). Concerning praxis, our proposed scheme refines the categorization of various EV brands, which plays an important role in the charging cost. Given the disparities in battery technology parameters among EVs, we add a coefficient $Brand$ (b) for synchronous V2V mode as a potential cost (when energy transfer occurs between EVs of different brands). (c) is mainly determined by energy transmission efficiency δ_ψ and transmission volume, where $\delta_{Syn} > \delta_{G2V} > \delta_{Asy}$ [8]. ρ is the original unit energy price.

IV. SCHEME DESIGN AND ALGORITHM

Fig.2 illustrates the computation logic of our proposed scheme. To minimize total charging costs, GC selects the charging mode with the lowest cost for each request generated by EV-C. Since our proposed scheme encompasses three charging modes simultaneously in the applicable scenario, Sections IV.A, IV.B, and IV.C correspond to the logic of synchronous V2V, asynchronous V2V, and G2V charging modes, respectively.

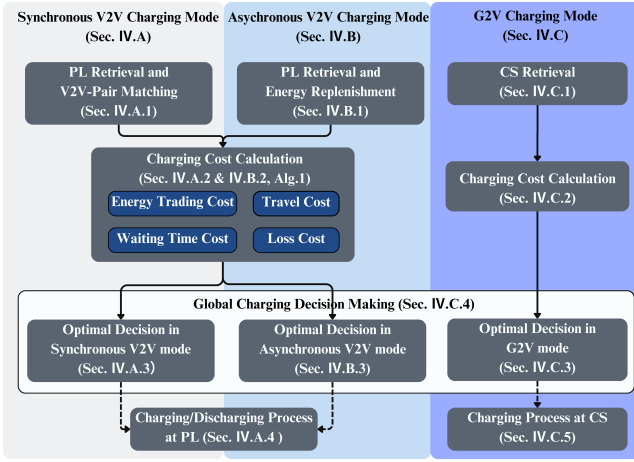


Fig. 2. Computation Logic of EV charging scheme

The logic is segmented according to the stages depicted in Fig.2, with specific implementation details presented in corresponding subsections. The charging location for G2V mode is CS, while synchronous V2V and asynchronous V2V mode share each PL. Therefore, differences arise in the execution logic across different charging modes, which encompass the selection of specific PL/CS, charging cost calculation and actual scheduling at local PL/CS. Algorithm 1 elaborates the calculation of charging cost at PL (for Γ_{Syn} , Γ_{Asy}), involving the integration details between synchronous and asynchronous V2V modes.

A. Synchronous V2V Charging Mode

When EV-C (denoted as EV- C_r)'s SoC drops below a certain threshold Thr_{evc} , it sends a charging request to GC. Through Sec.IV.A.1-IV.A.3, GC will generate an optimal decision in synchronous V2V mode for this charging request, which is formatted as $\{Syn \text{ (mode)}, PL_r \text{ (charging location)}, EV-P_r \text{ (matching counterpart)}, \Gamma_{Syn}\}$. Subscript r is used to denote the combination of entities. The optimal decision identified in Sec.IV.A.3 will serve as a candidate for this mode in the global mode selection discussed in Sec.IV.C.4. If this decision is chosen, it will guide EV- C_r to obtain actual charging service discussed in Sec.IV.A.4.

1) PL Retrieval and V2V-Pair Matching:

In this section, all possible charging locations and EV-Ps ('s' here represents the plural form of an entity and should be distinguished from subscripts) need to be retrieved and determined. GC first traverses all PLs in the scenario, generating a list $LIST_{pl}$ of all operational and accessible PLs. Each element in this list corresponds to a potential charging plan for EV- C_r , which is also formatted as $\{Syn, PL, EV-P, \Gamma_{Syn}\}$. The optimal decision is selected from all potential charging plans, hence the formats are the same. Given the spatial distribution variances of PLs, GC then selects different EV-Ps as matching counterparts for EV- C_r at different PLs. This predefined binding between PLs and EV-Ps ($\{PL, EV-P\}$) is aimed at reducing charging costs in subsequent charging cost calculation.

To minimize Γ_{Syn} , GC sets the combination of PL and EV-P based on three principles of V2V-pair matching: (a)

Distances from EV-C and EV-P to the PL should be similar, thereby reducing excessive waiting time for either party. Namely, minimizing the absolute difference in arrival times. (b) Prioritize matches between EVs of the same brand (*Brand*). (c) Subject to the first two principles, select EV-Ps that can reach the PL before EV- C_r . Priority is given to (a), followed by (b) if (a) cannot be fulfilled, and then (c). There are no conflicts among three principles, and they can be simultaneously satisfied. Through this section, GC constructs $LIST_{pl}$, with each PL binding a corresponding EV-P.

2) Charging Cost Calculation (Syn):

Before GC generates the optimal decision in synchronous V2V mode, it needs to compare all possible charging plans. In Sec.IV.A.1, the combinations of PLs and their corresponding EV-Ps have been formed. Each charging plan currently has three determined parameters $\{Syn, PL, EV-P\}$. In this section, GC further determines the final parameter Γ_{Syn} through Algorithm 1, which is used to evaluate each potential charging plan obtained in the first stage.

In Algorithm 1, GC iterates over each PL_i in $LIST_{pl}$ (corresponds to $\{Syn, PL_i, EV-P_i\}$) at line 1. We utilize subscript i to denote the binding relationship between PL_i and $EV-P_i$. Based on this entity combination, distances between EVs (EV- C_r , EV- P_i) and PL_i ($DIS_{pl(i)}^{evc(r)}$, $DIS_{pl(i)}^{evp(i)}$) are obtained through real-time position tracking. Thus travel energy consumption $E_{evc(r)}^{tra}$ is set as the result of the travel distance multiplied by the unit energy consumption P^{tra} .

GC sets EV- C_r 's energy demand $E_{evc(r)}^{req}$ through its charging request, while *Brand* depends on whether the brand of EV- C_r and EV- P_i is the same (line 2 in Algorithm 1). Based on distance and velocity V_{ev} , the expected arrival time of EV- C_r and EV- P_i at PL_i is then calculated (line 3). Since M^{Syn} (constant in synchronous V2V mode) is known, energy trading cost $COST_e$ can be calculated through equation (2) at line 4. Similarly, $COST_{tra}$ and $COST_{loss}$ are calculated based on known information, using equations (3) and (5).

Given the necessity for calculating $COST_{wait}$, Algorithm 1 needs further determine the waiting time $T_{pl(i)}^{wait}$ at PL_i in equation (4). Each PL/CS maintains N_{cha} and N_{wait} , respectively representing the lists of EVs currently charging/discharging and EVs queued up for charging/discharging at PL/CS. List TL is initialized with the completion times of charging/discharging processes for EVs in N_{cha} . It is sorted in ascending order, thus representing the timestamps when each parking space is released (line 5).

In Algorithm 1, N_{wait} is processed to predict $T_{pl(i)}^{wait}$ between lines 6-24. Since PL supports both synchronous and asynchronous V2V, EVs' charging modes need confirmation. If EV- k is in synchronous V2V mode, its expected charging duration T_{cha} is calculated (lines 6-8). From lines 9-17, depending on whether the arrival time of EV- k 's matching counterpart $T_{ev(k)}^{arr}$ is smaller than the time limit $TL[1]$. The first two elements of TL ($TL[0]$, $TL[1]$) may be updated as the completion time for this V2V-pair's energy transfer. Alternatively (the condition specified at line 13), EV- k is postponed by one position for processing.

Algorithm 1 Charging Cost Calculation

```
1: for each  $PL_i$  in  $LIST_{pl}$  do
2:   Set  $DIS_{pl(i)}^{evc(r)}$ ,  $DIS_{pl(i)}^{evp(i)}$ ,  $E_{evc(r)}^{req}$ ,  $E_{evc(r)}^{tra}$ ,  $Brand$ 
3:   Calculate the expected arrival time of EV- $C_r$  ( $T_{evc(r)}^{arr}$ ) and
   EV- $P_i$  ( $T_{evp(i)}^{arr}$ ) at  $PL_i$ .
4:   Calculate  $COST_e$ ,  $COST_{tra}$ ,  $COST_{loss}$ .
5:   Initialize  $TL$  with  $N_{cha}$  and sorted.
6:   for each  $EV_k$  in  $N_{wait}$  do
7:     if  $EV_k$  is in Syn V2V mode then
8:       Calculate  $T_{cha}$ .
9:       if  $EV_k$ 's matching counterpart is in  $N_{wait}$  then
10:         $TL[0] = TL[1] + T_{cha}$ ,  $TL[1] = TL[0]$ 
11:       else
12:        Ensure the arrival time of its matching counterpart
        ( $T_{ev(k)}^{arr}$ ).
13:        if  $TL[1] < T_{ev(k)}^{arr}$  then
14:          Delay  $EV_k$ 's processing by one position.
15:        else
16:          Execute the same operation as in line 10.
17:        end if
18:      end if
19:      else if  $EV_k$  is in Asy V2V mode then
20:        Calculate  $T_{cha}$  or  $T_{dis}$ .
21:         $TL[0] = TL[0] + T_{cha}(T_{dis})$ 
22:      end if
23:      Sort  $TL$  in ascending order.
24:    end for
25:    if  $MAX(T_{evc(r)}^{arr}, T_{evp(i)}^{arr}) < TL[1]$  then
26:       $T_{pl(i)}^{wait} = TL[1] - T_{evc(r)}^{arr}$ 
27:    else
28:       $T_{pl(i)}^{wait} = MAX(T_{evc(r)}^{arr}, T_{evp(i)}^{arr}) - T_{evc(r)}^{arr}$ 
29:    end if
30:    Calculate  $COST_{wait}$  via equation (4), then  $\Gamma_{Syn(i)}$  via
    equation (1).
31:    Record charging plan  $\{Syn, PL_i, EV-P_i, \Gamma_{Syn(i)}\}$ .
32:  end for
33: return All possible charging plans.
```

If EV_k is in asynchronous V2V mode, the anticipated charging/discharging duration T_{cha}/T_{dis} for EV_k needs to be calculated. $TL[0]$ is directly updated to its completion time for charging/discharging between lines 19-22. TL is then re-sorted, ensuring that the release time of charging spaces is always arranged in chronological order (line 23). $MAX(T_{evc(r)}^{arr}, T_{evp(i)}^{arr})$, the estimated start time for EV- C_r and EV- P_i to begin energy transfer is compared with $TL[1]$. The larger one is subtracted from $T_{evc(r)}^{arr}$ as $T_{pl(i)}^{wait}$ (lines 25-29).

At this stage, $COST_{wait}$ is calculated via equation (4). Then $\Gamma_{Syn(i)}$ corresponding to PL_i is obtained using equation (1) at line 30 in Algorithm 1. Line 32 illustrates the composition of a complete potential charging plan, while Algorithm 1 ultimately returns the set of all charging plans corresponding to $LIST_{pl}$ (line 33).

3) Optimal Decision in Synchronous V2V Mode:

Each feasible charging plan returned by Algorithm 1 now encompasses all parameters, they will be further compared in this section. $\{Syn, PL_i, EV-P_i, \Gamma_{Syn(i)}\}$ signifies that EV- C_r can travel to PL_i and engages in V2V energy transfer with EV- P_i , at the cost of $\Gamma_{Syn(i)}$. To minimize the global charging cost, GC will select the plan with the lowest cost

as the ultimate optimal decision in synchronous V2V mode, which is represented as $\{Syn, PL_r, EV-P_r, \Gamma_{Syn}\}$. It will be further compared with the optimal decisions in the other two modes in Sec.IV.C.4.

4) Charging/Discharging Process at PL:

If GC selects the optimal decision in synchronous V2V mode in the global mode selection (Sec.IV.C.4), EV- C_r will adhere to this charging mode and head to the designated PL_r to obtain energy from EV- P_r . Before getting charged, EV- C_r will first be added to the waiting queue N_{wait} . This section focuses on the scheduling issue within PL, addressing how EVs are scheduled from the waiting queue for charging/discharging.

Since both synchronous and asynchronous V2V modes share parking space allocation at PL, EV scheduling needs to simultaneously consider the availability of parking spaces and energy reserves. For a waiting EV in synchronous V2V mode, the estimated arrival time of its matching counterpart is crucial. When two parking spaces become available, V2V-pair will use them based on the waiting sequence, which follows the logic similar to lines 9-17 in Algorithm 1. For a waiting EV in asynchronous V2V mode, it just needs one parking space. When the energy reserve of PL falls below a threshold (Thr_{pl}), it will prioritize scheduling asynchronous EV-Ps to ensure the normal functioning of asynchronous V2V mode.

The primary principle of scheduling at PL is based on First Come First Served (FCFS). Additionally, we consider the existence of specified EVs (police car, ambulance, etc). Given the primary principle, the secondary principle dictates that they are prioritized for scheduling from the waiting queue compared to ordinary EVs. When EVs obtain parking spaces, they will undergo charging/discharging processes based on their modes and roles (EV-C/EV-P).

B. Asynchronous V2V Charging Mode

In response to the charging request from EV- C_r , GC will derive the optimal decision in asynchronous V2V charging mode through Sec.IV.B.1-IV.B.3. It, as well as potential charging plans, are formatted as $\{Asy, PL, \Gamma_{Asy}\}$. Subsections outline the general process and delve into differences from synchronous V2V mode.

1) PL Retrieval and Energy Replenishment:

Similar to $LIST_{pl}$ in Sec.IV.A.1, GC also generates the set $LIST_{pl}^{Asy}$ of all PLs supporting asynchronous charging. The distinction between these two lists ($LIST_{pl}$ and $LIST_{pl}^{Asy}$) lies in an additional prerequisite: sufficient energy reserves. This entails estimating the energy consumption of EV-Cs and the energy discharge of EV-Ps in asynchronous mode at PL, thereby determining if a PL can support asynchronous charging. PLs with SoC below Thr_{pl} send energy replenishment requests, prompting GC to arrange EV-Ps to provide asynchronous discharge based on distance.

2) Charging Cost Calculation (Asy):

GC traverses $LIST_{pl}^{Asy}$ and evaluates the impact of each feasible PL on the cumulative global charging cost. Calculation of Γ_{Asy} follows the equations (1)-(5). All potential

charging plans are generated through Algorithm 1 as well, but certain modifications are required. EV- P_i does not exist, calculations related to it can be omitted. The logic for predicting waiting time is the same as lines 5-24, except that after obtaining the final TL , there is no need to consider the matching counterpart. $TL[0]$, the time when the first charging space is released after all waiting EVs commence charging/discharging, is compared with $T_{evc(r)}^{arr}$. The difference between the larger one and $T_{evc(r)}^{arr}$ becomes $T_{pl(i)}^{wait}$.

3) Optimal Decision in Asynchronous V2V Mode:

In this section, GC compares all possible charging plans and select the one with the minimum charging cost as the optimal decision. It is represented as $\{Asy, PL_r, \Gamma_{Asy}\}$. Subsequently, it will be proposed for mode selection in Sec.IV.C.4. If the optimal decision in asynchronous V2V mode is selected, GC will notify EV- C_r of the designated PL_r . Once confirmed by EV- C_r , it will proceed to obtain energy from the deployed energy storage facility at PL_r .

C. G2V Charging Mode

GC gets the optimal decision in G2V mode through Sec.IV.C.1-IV.C.3. The optimal decision and potential charging plans are formatted as $\{G2V, CS, \Gamma_{G2V}\}$. Sec.IV.C.4 describes the logic of global mode selection.

1) CS Retrieval:

In G2V charging mode, GC generates $LIST_{cs}$, which contains all operational and accessible CSs for EV- C_r . In practice, the number of CSs in the scenario is fewer than PLs, and they are relatively distributed closer to the central area.

2) Charging Cost Calculation (G2V):

In this section, all complete charging plans are generated. The calculation logic of charging cost in G2V mode differs from the other two modes. Firstly, G2V adopts a ToU pricing model given by [16], where the unit energy price M_t^{G2V} fluctuates with the actual grid load. Unlike line 4 in Algorithm 1, $COST_e$ (equation (2)) needs to be calculated after the waiting time at CS (T_{cs}^{wait}) is predicted. As the subscript t of M_t^{G2V} corresponds to the anticipated start time of charging. Secondly, the logic for predicting T_{cs}^{wait} differs from Algorithm 1, as CS only supports G2V mode. Nothing but EV-Cs in G2V mode appear in CS, making the mode judgment or V2V-pair arrival assessment in Algorithm 1 meaningless. The time when the first charging slot is released after all EV-Cs in the waiting queue commence charging can determine T_{cs}^{wait} .

3) Optimal Decision in G2V Mode:

Each CS in $LIST_{cs}$ corresponds to a potential charging plan. The plan with the minimum charging cost is selected as the optimal decision, represented as $\{G2V, CS_r, \Gamma_{G2V}\}$, in G2V mode, thus participating in the inter-mode comparison in Sec.IV.C.4.

4) Global Charging Decision Making:

The optimal decisions for three charging modes are respectively determined in Sec.IV.A.3, IV.B.3, IV.C.3. GC will compare these three decisions and choose the one with the lowest cost (among Γ_{Syn} , Γ_{Asy} , Γ_{G2V}) as the final decision.

EV- C_r will be informed by GC, and it will proceed to receive charging service at PL_r (Sec.IV.A.4) or CS_r (Sec.IV.C.5).

5) Charging Process at CS:

Within CS, only the charging process of EV-Cs exists, without discharging process. The scheduling issue at CS doesn't necessitate considering the arrival of V2V-pairs and PLs' energy storage constraints. Similar to Sec.IV.A.4, EV scheduling adheres to FCFS principle, granting precedence to specialized EVs for service thereafter. After an EV-C finishes charging, the vacated charging slot is then utilized by a waiting EV-C in the waiting queue.

V. EXPERIMENTAL SETUP

We have built up an entire hybrid charging system based on Opportunistic Network Environment (ONE) to simulate the proposed scheme. Fig.3 illustrates the simulation scenario, which represents the urban area of Helsinki city, Finland, with dimensions 4500x3400 square meters. Within this urban area, 32 PLs and 12 CSs are geographically distributed. Each PL provides 12 parking spaces along with an energy storage facility. EVs in synchronous V2V mode utilize onboard DC-DC converters for energy transfer, operating at a rate of 30 kW. Charging/Discharging powers for EVs in asynchronous mode are set to 40 kW. Each CS offers 5 charging slots and supports a highest charging rate of 60 kW.

A. Simulation Configuration

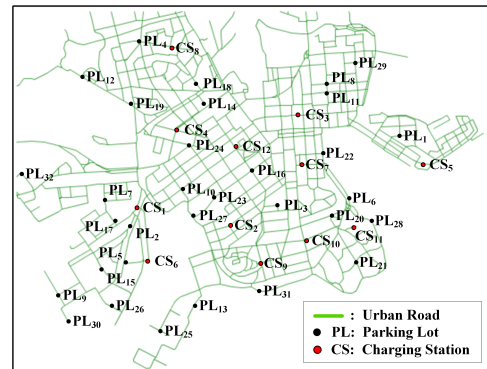


Fig. 3. Simulation Scenario

Meanwhile, the unit energy price in synchronous mode (M^{Syn}) is set to 10 cents/kWh [17], while M^{Asy} is set to 12 cents/kWh. M_t^{G2V} is adjusted based on the grid load within the range of [12, 20] [16].

To simulate uniformly occurring charging demands, EV-Cs of each type are divided into five groups based on initial SoC (60% to 100%). Similarly, EV-Ps are divided into three types ($Brand_{1-3}$), with an initial energy of 60 kWh and a 30% SoC threshold. To further simulate real-world scenario, 5% of EVs are randomly set as specialized EVs with high-service priority. The simulation duration is 12 hours (6.00-18.00, summer weekday), and entity information is updated at a frequency of 0.1s.

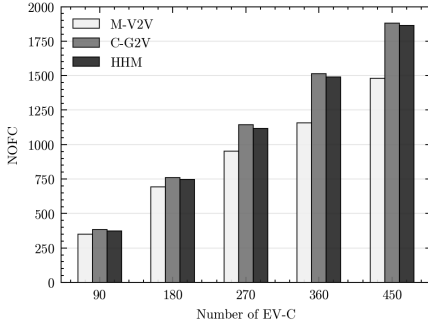


Fig. 4. NOFC

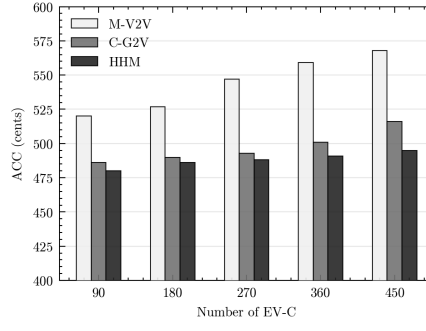


Fig. 5. ACC

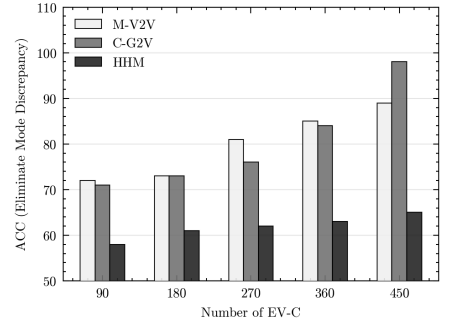


Fig. 6. ACC (Eliminate Mode Discrepancy)

B. Comparison Configuration

The charging schemes used for comparison are as follows:

MWT-based V2V Charging (M-V2V) [15]: This combines synchronous with asynchronous V2V mode, pairs V2V based on the shortest distance, and selects charging mode based on Minimum Waiting Time (MWT).

CC-based G2V Charging Mode (C-G2V): To simulate real-world scenarios, it is separated from HHM. It selects CS for G2V charging based on Charging Cost (CC).

HHM-based Charging Scheme (HHM): The proposed scheme integrates three modes, which makes decisions based on charging cost (V2V-pair matching, charging mode selecting and charging location designating).

Metrics used for performance evaluation are as follows:

Number of EVs Fully Charged (NOFC): This metric represents the total number of EVs that get fully charged under each mode during the simulation period. It intuitively reflects the effectiveness of the scheme.

Average Charging Cost (ACC): It represents the average charging cost incurred during a single charging process. It is directly related to user's QoE, thus reflecting the decision quality of the scheme.

C. Charging Scheme Performance Evaluation

In this section, we evaluate the performance of each scheme by adjusting the number of EV-C in the scenario (ranges from 90 to 450), while the number of EV-Ps remains fixed at 180. EVs travel at a constant speed. Among them, EVs are divided into three types:

- Coda Automotive ($Brand_1$) {33.8 kWh (battery capacity), 193 km (traveling range), 30% (SoC threshold)}
- Wheego Whip ($Brand_2$) {30 kWh, 161 km, 40%}
- Hyundai BlueOn ($Brand_3$) {16.4 kWh, 140 km, 50%}

Fig.4-6 present comparison results of three schemes based on the metrics used for performance evaluation.

From Fig.4, M-V2V exhibits the poorest performance in terms of service provision quantity, as the nearly twofold gap in charging rates between V2V and G2V mode is difficult to bridge. HHM demonstrates high stability and displays the capability to address surges in charging demand, as it can utilize heterogeneous charging facilities to allocate demand, thus ensuring high NOFC. Meanwhile, C-G2V benefits from its superior charging rate, resulting in slightly higher NOFC than HHM.

From Fig.5, it's evident that HHM attains the lowest ACC, while its increase remains the most consistent as the number of EV-C grows. Though C-G2V exhibits a NOFC similar to HHM in Fig.4, charging congestion (attributed to excessive charging demand) at CSs leads to higher ACC than HHM. The gap widens as the number of EV-C increases. M-V2V, by solely focusing on waiting time and overlooking other cost factors, results in the highest ACC.

Considering the discrepancies in charging rate and the energy price among three modes, ACC (Eliminate Mode Discrepancy) is computed in Fig.6. Factors constrained by the mode itself are removed, while travel time, waiting time, and loss cost are retained. The adjusted ACC is equally effective in evaluating the performance of schemes. Compared to Fig.5, the advantage of HHM in reducing charging cost is more pronounced. C-G2V experiences rapid increases in waiting time due to charging congestion, while M-V2V continues to overlook the loss cost.

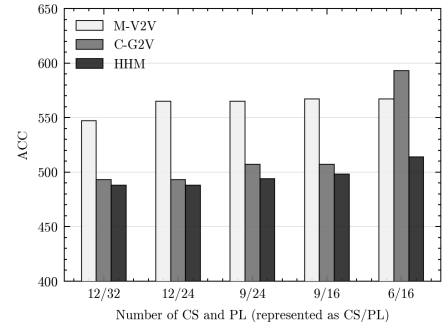


Fig. 7. Influence of CS and PL deployment

To further demonstrate the broad applicability and robustness of the proposed scheme, we fix the number of EV-C at 270 and vary those of CS and PL (adjusting the ratio). As shown in Fig.7, HHM consistently maintains the lowest ACC as the number of charging sites decreases. Though M-V2V maintains a relatively stable ACC at the expense of service quantity, it still exhibits the highest ACC, indicating it cannot fully replace the G2V mode. C-G2V shows a significant cost increase when the number of CS is halved, primarily due to $COST_{tra}$ incurred by geographical distribution constraints and charging congestion caused by demand exceeding service capacity. Fig.7 illustrates that our scheme can effectively address the real-world scenarios of energy supply and demand mismatch.

D. Time Distribution of Mode Selection

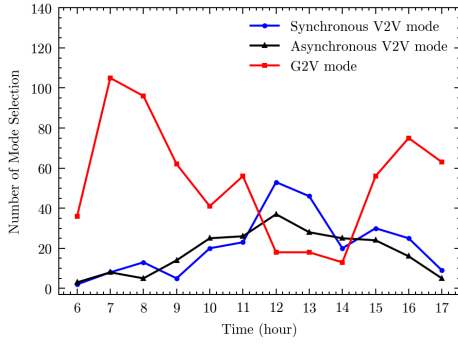


Fig. 8. Selection frequency of each mode

Fig.8 illustrates the comparison of mode selection results during the daytime (270 EV-C, HHM). In initial period (6.00-11.00), energy demands begin to emerge. G2V mode is prioritized due to its highest charging rate and low ToU price. The decreasing trend reflects the charging congestion at CSs, causing an increase in selection frequency of V2V modes.

During midday period (11.00-14.00), which is a peak energy consumption period, decisions shift towards both V2V modes. In final period (14.00-18.00), the selection frequency of G2V mode rebounds as the grid load is alleviated. Due to energy storage constraints at PL, the trend for the asynchronous mode is relatively stable, with fewer selections compared to synchronous mode.

Our proposed scheme can efficiently leverage heterogeneous charging facilities in the scenario, utilizing the hybrid charging modes to transfer charging demands, thereby balancing the grid load.

VI. CONCLUSION

To meet the increasing charging demand of EVs, we proposed an urban EV charging scheme composed of three charging modes (synchronous V2V, asynchronous V2V and G2V). The scheme fully utilized heterogeneous charging facilities in urban settings to achieve hybrid charging. Additionally, we introduced a charging cost model from EV-C side. Following the principle of minimizing costs, GC selected charging modes and delineated corresponding charging plans. This paper further employed the urban topology of Helsinki to compare three charging schemes. The result showed that our proposed scheme ensured quantity of services provided, while achieving the lowest ACC. This contributed to load balancing in power system and user's QoE enhancing.

In the future work, we will adopt fully or partially distributed scheduling to replace the centralized GC, thereby reducing computational overhead. The current scheme doesn't prioritize the decision-making efficiency of GC when facing large-scale energy demand. Additionally, we plan to extend the cost model to encompass the entire system, including the computational cost of GC and various costs incurred by EV-P. Finally, interactions among energy suppliers, which our current work doesn't address, are also present. Considering

practical applications, it's necessary to design a profitable business model for CS and EV-P as economic entities.

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